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SEASONAL FLUCTUATION IN $\delta^{15}N$ OF GROUNDWATER NITRATE IN A MANTLED KARST AQUIFER DUE TO MACROPORE TRANSPORT OF FERTILIZER-DERIVED NITRATE

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ABSTRACT

Wells, E.R. and Krothe, N.C., 1989. Seasonal fluctuation of $\delta^{15}N$ of groundwater nitrate in a mantled karst aquifer due to macropore transport of fertilizer-derived nitrate. J. Hydrol., 112: 191–201.

Twenty wells were sampled and analyzed for nitrate content and nitrogen isotopes during late September, 1986. The groundwater nitrate $\delta^{15}N$ values ranged from +6.8 to +17.6‰ with a mean of +10.3‰. Eleven samples had $\delta^{15}N$ values greater than +9.5‰ indicating an animal waste origin. Septic tanks are probably the predominant source of waste-derived nitrate. Macropores draining septic tank filter fields may provide a nearly continuous flow of effluent to the aquifer. The $\delta^{15}N$ values of the other nine samples ranged from +6.8 to 8.4‰ suggesting a possible mixing of waste-derived nitrate and that derived from isotopically lighter cultivation sources, i.e. inorganic fertilizer and/or mineralized soil organic nitrogen.

The twenty wells were resampled during late May, 1987 shortly after inorganic fertilizers had been applied to the corpland throughout the study area. The groundwater nitrate $\delta^{15}N$ values ranged from +3.6 to +14.7‰ with a mean of +8.3‰. Nineteen of the wells showed a shift towards the lighter $\delta^{15}N$ values expected from fertilizer-derived nitrate. The $\delta^{15}N$ decreases ranged from 0.6 to 4.8‰ with a mean of 2.2‰. Despite the apparent influx of fertilizer-derived nitrate, eleven samples (with $\delta^{15}N$ +8.0 to +14.7‰) still were composed of predominantly waste-derived nitrate. Only nine samples (with $\delta^{15}N$ +3.6 to +5.8‰) may have had a fertilizer-derived component approaching in magnitude that of waste-derived nitrate. Regional groundwater nitrate concentration did not substantially increase

Macropore (cracks, root channels, etc.) flow appears to contribute significantly to the recharge of the aquifer. Rapid response of the groundwater nitrate $\delta^{15}N$ values to fertilization demonstrates the ability of macropore flow to transport fertilizer-derived nitrate which has been flushed from the tilled soil layer. Other agriculture chemicals (pesticides and herbicides) may also be transported to the aquifer in this manner.

INTRODUCTION

Groundwater nitrate (NO_3^-) pollution occurs worldwide. In many regions crop cultivation and disposal of sewage and animal wastes are potential sources of nitrate. Transport of the nitrate from source to aquifer depends on many hydrologic and geologic factors. As a result, the primary origin of the groundwater nitrate is undetermined in most cases.

A groundwater survey showed that nitrate (NO_3^-) is a common contaminant of a mantled portion of a karst aquifer in southern Indiana (Wells, 1988). At numerous locations nitrate concentration exceeds the recommended drinking-water limit of 45 mg l^{-1}. The mantled aquifer underlies an agricultural region in which septic tank effluents, livestock wastes, inorganic fertilizers, and mineralized soil organic nitrogen are potential sources of nitrate.

Flow through macropores in the mantle probably provides a substantial portion of the local groundwater recharge. It may be significant in the nitrate contamination by directly flushing of septic tank effluents and fertilizers to the aquifer.

Kreitler (1975), Gormly and Spalding (1979), and Flipse and Bonner (1985), among many others, have shown that semiquantitative distinction between groundwater nitrate derived from human or animal waste and that derived from cultivation sources, i.e. inorganic fertilizer or mineralized soil organic nitrogen, is possible on the basis of its $^{15}N/^{14}N$ ratio. Waste-derived nitrate is generally enriched in ^{15}N relative to cultivation-derived nitrate. Heaton (1986) presents a detailed review of isotopic investigations concerning nitrogen pollution of the hydrosphere.

An isotopic study of the nitrate-nitrogen from the waters of the southern Indiana mantled karst aquifer was undertaken during the fall of 1986 and the spring of 1987. The purpose of the study was twofold: (1) to determine the predominant source(s) of the groundwater nitrate; and (2) to assess the role of macropore recharge in the contamination. Special consideration was given to the possibility of short-term, seasonal flushing of fertilizer-derived nitrate to the aquifer. The results of that study are examined in this paper.

STUDY AREA

Physiography and hydrogeology

The study area is located on the Mitchell Plain of southern Indiana (Fig. 1). The Mitchell Plain is a physiographic province that has developed on a sequence (~120 m thick) of gently dipping (6 m km^{-1} to the southwest), well fractured, Mississippian limestones which have undergone extensive karstification (Palmer and Palmer, 1975). The limestones are the sole source of groundwater to much of the plain's rural population.

The majority of the plain is characterized by undulating karst topography with many sinkholes, sinking streams, and springs. However, smaller, discontinuous portions of the plain are covered by a clayey mantle that disguises the karst nature of the underlying bedrock. The mantle is composed of residuum from the argillaceous limestone and alluvium formed during the westward retreat of the neighboring Crawford Upland (Fig. 1). The sedimentary mixture was homogenized by weathering into a thick, clayey soil. The soil was subsequently capped by a thin (< 2 m) Wisconsinan loess. The loess and paleosol are commonly separated by a lag concentrate of chert pebbles and cobbles

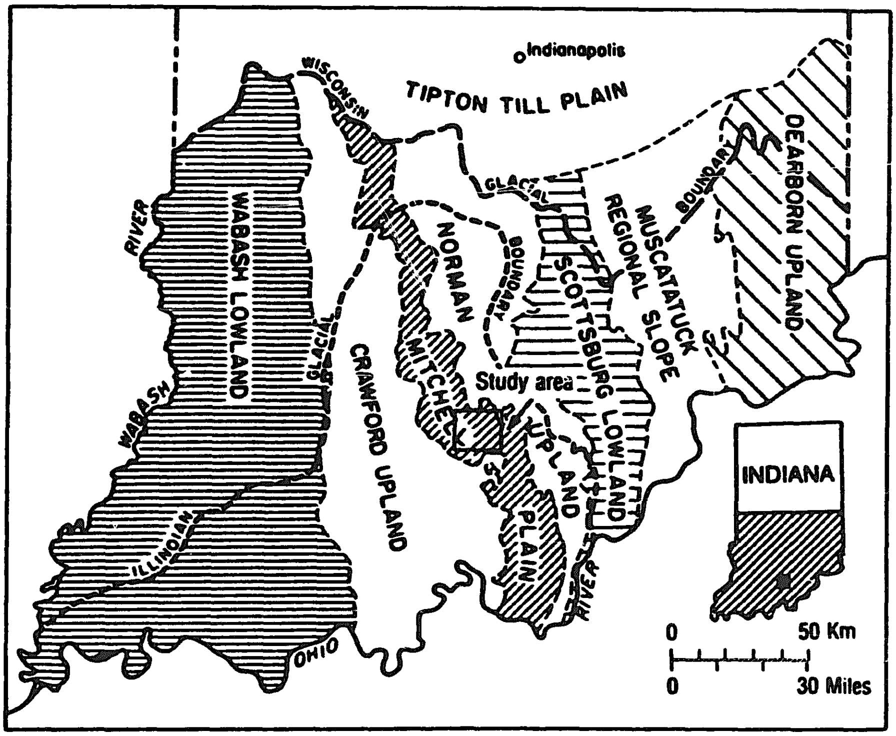

Fig. 1. Map of southern Indiana showing location of study area and physiographic provinces (modified from Sunderman, 1968).

indicating of an erosion surface that predates the loess. The clayey paleosol has often been termed "terra rossa" or "red earth" (Ruhe, 1975).

Within the study area, unconsolidated sediments are as much as 24 m thick, although average about 10 m (Ruhe, 1975). The resulting landscape is flat to gently rolling with very few observable karst features, i.e. large sinkholes. A surface drainage network exists through the two forks of the Lost River (Fig. 2) and its tributary streams.

Recharge to the karst aquifer is locally derived from precipitation. Despite the presence of the clayey mantle, groundwater flow in the limestone responds rapidly to rainfall in typical karst fashion. Muddy water is pumped from many wells shortly after rainstorms. Substantial recharge probably occurs via macropore flow through the mantle. Macropores are cracks, fissures, root channels, animal burrows, and textural boundaries in the unsaturated zone of the sediment in which water is subjected to minimal capillary force. Macropore flow is commonly several orders of magnitude more rapid than that in the surrounding sedimentary matrix (Quinlan and Aley, 1987).

The importance of macropore flow in karst regions has been noted by others, e.g. Thomas and Phillips (1979). Aley (1977) has estimated that in a karst area of the Missouri Ozarks water entering soil macropores contributes over four times as much to groundwater recharge than does diffuse flow through the sedimentary matrix. Although no attempt has been made to quantify the macropore or diffuse recharge to the karst aquifer in this study, macropore recharge probably is predominant.

The groundwater occurs in the limestones of the Sanders and overlying Blue River Groups. Significant flow is restricted to solution cavities formed along fractures, joints, and bedding planes. The diameters of some solution cavities are measured in meters. Most wells are cased through unconsolidated sediment and are finished in the thinly bedded Blue River Group limestones which have undergone the most extensive dissolution. Well yields vary according to the number and size of solution cavities intersected, but are more than sufficient for domestic and farm supply. Few wells exceed 45 m in depth.

Land use and soils

Land use is primarily agricultural. The majority (> 65%) of the land is used for the production of corn, soybeans, and wheat. Inorganic nitrogenous fertilizers are applied to this cropland at a rate of 170–200 kg N ha^{-1} yr^{-1} (B.W. Fagg, pers. commun., 1987). Anhydrous ammonia is the predominate type, but ammonium nitrate and urea are also commonly used. Manure is spread on fields where available. The region receives an average of 114 cm of precipitation annually, therefore irrigation is unnecessary.

The excessively slopig and otherwise nonarable lands along major drainage ways are generally used for pasture. Livestock and poultry are raised on many

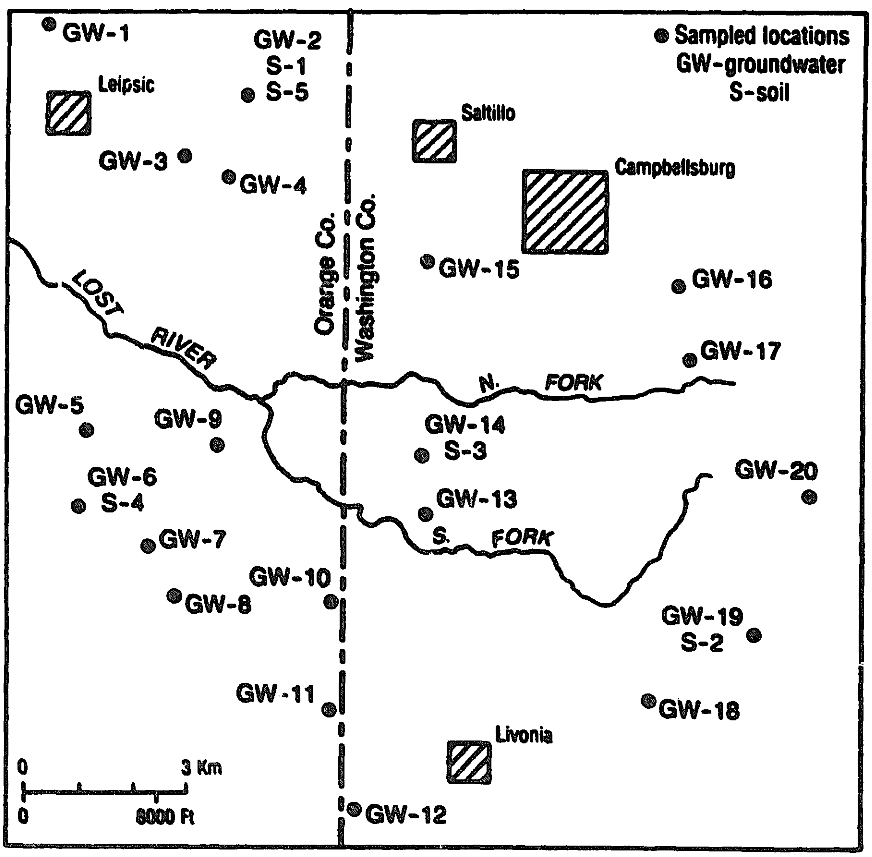

Fig. 2. Map of study area showing location of sampled wells and soils.

farms. Four very small towns are located in the study area (Fig. 2). Campbellsburg (population 695), the largest, is the only one to have water and sewer systems. The untreated sewage is collected in lagoons just northwest of the town. Waste disposal throughout the rest of the study area is accomplished by use of septic systems.

The soils are silt loams formed in the thin loess and in the underlying red earth. Clay content greatly increases with depth; red and brown clays constitute the lower portions of subsoils. The subsoils typically extend to depths of 200 cm or more. Soil acidity ranges from extremely acidic (pH < 4.5) to neutral (pH 6.6–7.3) (Miller et al., 1939; Wingard, 1984).

SAMPLE COLLECTION AND ANALYSIS

Water samples were obtained from twenty domestic and farm wells (Fig. 2) representing a wide range of nitrate concentration (Wells, 1988). The samples were collected in 2-l polyethylene bottles and placed on ice. Upon return to the laboratory the samples were immediately filtered, placed in 1-l polyethylene bottles, and frozen. The samples were later analyzed for nitrate concentration and nitrogen isotopic composition. The sampling took place in late September, 1986 and in late May, 1987. Nitrate concentration was determined by a slightly modified version of the Bremner and Keeney (1965) steam distillation procedure. The modification was that of Gormly and Spalding (1979). The procedure produced a titrated and concentrated ammonium sulfate solution which was preserved for subsequent nitrogen purification and isotopic analysis.

The ammonium sulfate was converted to nitrogen gas with sodium hypobromite and purified according to Bremner and Edwards (1965) and Miyaka and Wada (1967). The $^{15}N/^{14}N$ ratios of the purified nitrogen gas were determined on a Nuclide^{R} 6-60 isotope-ratio mass spectrometer. Atmospheric nitrogen, prepared by the procedure of Cline (1973), was the primary working standard. The isotopic ratios were reported in standard notation as:

$$\delta^{15}N\ (‰) = [(R_{sample}/R_{standard}) - 1] \times 10^3$$

where $R = {}^{15}N/^{14}N$. Total experimental error was less than 0.5‰.

A hand auger was used to collect soil samples at five locations (Fig. 2) during early June, 1987. The depths of 0–10 cm and 30–40 cm were sampled. The samples were sealed in plastic bags, frozen, and stored for analysis.

Leachable ammonia and nitrate were recovered from the sediment with 2N KCl solution in a wrist-action shaker operated for 1 h. A 1:10 sample to leachate solution ratio was used to ensure complete leaching (Lindau and Spalding, 1984). Where several extractions were required because of low nitrate concentration, the extracts were concentrated by rotary evaporation. The extracted ammonia and nitrate were analyzed by the previously described procedures for water samples.

RESULTS AND DISCUSSION

Groundwater sampled in September, 1986

The nitrate concentrations and nitrogen isotopic compositions of the groundwater nitrate are presented in Table 1. The $\delta^{15}N$ values ranged from + 6.8 to + 17.6‰ and averaged + 10.3‰.

The large degree of ^{15}N enrichment observed in many of the samples indicates that animal wastes are an important source of nitrate. Decomposing animal wastes (original $\delta^{15}N$ = + 2 to + 5‰; Gormly and Spalding, 1979) become enriched in ^{15}N when ammonia, strongly depleted in the heavier isotope, is lost through volatilization. The nitrate subsequently produced from the enriched residual waste material typically has a $\delta^{15}N$ value of between + 9 and + 22‰ (Kreitler, 1975; Gormly and Spalding, 1979; Lindau and Spalding, 1984).

Eleven of the twenty samples had isotopic compositions (+ 9.6 to + 17.6‰) within the range for nitrate produce from animal waste. Throughout the investigated area septic tanks are probably responsible for the majority of the

TABLE 1

Concentrations and nitrogen isotopic compositions of nitrate from groundwater samples

Sample	NO_3^- (mg l^{-1})			$\delta^{15}N$ (‰)		
	Sept. 1986	May 1987	Change*	Sept. '86	May '87	Change*
GW-1	5.3	8.4	+ 3.1	+ 8.2	+ 10.6	+ 2.4
GW-2	75	75	0	+ 15.6	+ 14.7	- 0.9
GW-3	49	53	+ 4	+ 11.1	+ 10.2	- 0.9
GW-4	31	39	+ 8	+ 6.8	+ 4.2	- 2.6
GW-5	4.9	8.0	+ 3.1	+ 10.2	+ 8.0	- 2.2
GW-6	37	44	+ 7	+ 14.1	+ 12.8	- 1.3
GW-7	53	36	- 17	+ 10.4	+ 5.6	- 4.8
GW-8	13	11	- 2	+ 14.3	+ 13.0	- 1.3
GW-9	8.0	8.9	+ 0.9	+ 7.4	+ 5.4	- 2.0
GW-10	35	44	+ 9	+ 17.6	+ 14.0	- 3.6
GW-11	49	49	0	+ 7.6	+ 4.2	- 3.4
GW-12	37	35	- 2	+ 6.9	+ 5.4	- 1.5
GW-13	35	44	+ 9	+ 8.4	+ 4.2	- 4.2
GW-14	12	26	+ 14	+ 6.8	+ 4.9	- 1.9
GW-15	19	21	+ 2	+ 10.8	+ 9.4	- 1.4
GW-16	22	26	+ 4	+ 8.4	+ 3.6	- 4.8
GW-17	16	16	0	+ 7.4	+ 5.8	- 1.6
GW-18	26	27	+ 1	+ 14.8	+ 13.3	- 1.5
GW-19	128	133	+ 5	+ 9.6	+ 8.3	- 1.3
GW-20	22	30	+ 8	+ 9.9	+ 9.3	- 0.6
Average	34	37		+ 10.3	+ 8.3	

* Change in value from September '86 to May '87.

waste-derived groundwater nitrate as they are the most common type of animal waste source and are often located close to wells. Other animal waste sources such as barnyards, feedlots, pastures, etc. are undoubtedly locally important in the contamination of certain wells. Nitrate from any of these waste sources can be rapidly flushed to the groundwater by storm runoff entering macropores. Septic tank filter fields may be drained through macropore flow and thereby provide a nearly continuous supply of effluent to the aquifer.

The $\delta^{15}N$ values of the other nine samples fell within the narrow range of +6.8 to 8.4‰. These values probably result from a mixture of waste-derived nitrate and that derived from isotopically lighter cultivation sources, i.e. inorganic fertilizer and/or mineralized soil organic nitrogen. Within the study area, nitrate derived from cultivation most likely has a $\delta^{15}N$ value of less than +6‰ (this report).

Alternatively, these samples less enriched in ^{15}N may have been derived solely from animal wastes. Occasionally, waste sources produce unusually light ($\delta^{15}N$ < +9‰) nitrate (Wolterink et al., 1979; Lindau and Spalding, 1984). This probably occurs when only a very small amount of ammonia is volatilized during waste decomposition. Wolterink et al. (1979) found that nitrate derived from septic tank effluent had a lower average $\delta^{15}N$ value (+10.9‰) and was more frequently below +9‰ than that derived from feedlots and barnyards (average +12.4‰). Apparently, in some septic tanks and filter fields the conditions of low temperature, high moisture, and little air movement minimize ammonia volatilization and therefore produce the poorly enriched nitrate. Perhaps this same situation exists in some of the many septic systems in the study area.

Groundwater sampled in May, 1987

During April and May inorganic nitrogenous fertilizers were applied to fields throughout the study area. Work by Quisenberry and Phillips (1976, 1978) had shown that even in tilled soils, where macropores had been disrupted to a depth of 15 cm, deep macropore flow still occurred, though to a lesser extent than in undisturbed soil. Therefore, the potential existed for the flushing of fertilizer to the karst aquifer via macropores beneath the fields. In order to detect any influx of fertilizer-derived nitrate, the wells were sampled in late May. The $\delta^{15}N$ values ranged from +3.6 to +14.7‰ and averaged +8.3‰ (Table 1). Nineteen of the twenty wells showed a decrease (averaging 2.2‰) in $\delta^{15}N$ value from the previous September sampling (Fig. 3, Table 1). Only well GW-1 showed an increase. This general decline in groundwater nitrate $\delta^{15}N$ value probably is attrributable to isotopically light fertilizer-derived nitrate flushed to the karst aquifer through macropore flow.

Inorganic fertilizers have $\delta^{15}N$ values generally ranging between −6 and +7‰, but nearly 90% of the samples have values of less than +3‰ (Heaton, 1986). Anhydrous ammonia, the predominant fertilizer type in the study area, typically has a $\delta^{15}N$ value of less than 0‰ (Spalding et al., 1982). Upon applica-

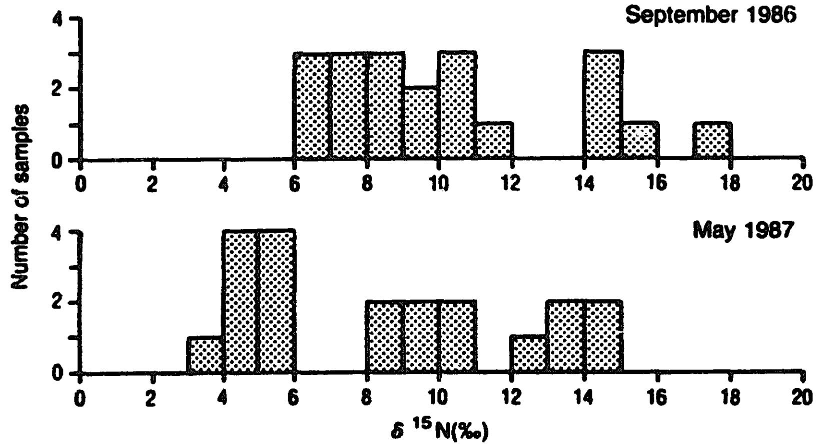

Fig. 3. Histograms of groundwater nitrate $\delta^{15}N$ values from September, 1986 and May, 1987.

tion to the soil, the fertilizer nitrogen, if in reduced form, is nitrified to nitrate. The nitrate retains an isotopic composition very similar to that of the original fertilizer if little or no volatilization of ^{15}N-depleted ammonia occurs before the nitrification process goes to completion. Soil pH is an important factor controlling ammonia volatilization; alkaline conditions promote volatilization and can result in fertilizer- derived nitrate being enriched in ^{15}N (Fenn and Kissel, 1973; Kreitler, 1979). The acidic soils within the study area should have

TABLE 2

Concentrations and nitrogen isotopic compositions of leachable nitrate and ammonia from soil samples

Sample*	NH^3		NO_3^-		Remarks
depth (cm)	$\mu g\, g^{-1}$	$\delta^{15}N$ (‰)	$\mu g\, g^{-1}$	$\delta^{15}N$ (‰)	soil, utilization, fertilizer applied
S-1					Bromer silt loam,
0–10	< 12	–	456	+ 0.1	corn, anhydrous NH_3
30–40	< 12	–	49	–	
S-2					Frederick silt
0–10	109	+ 15.1	704	– 9.8	loam, corn,
30–40	< 12	–	89	– 5.7	anhydrous NH_3
S-3					Bedford silt
0–10	< 12	–	474	– 4.0	loam, corn,
30–40	< 12	–	22	–	undetermined
S-4					Bedford silt
0–40	< 12	–	177	+ 8.8	loam, veg. garden,
30–40	< 12	–	168	+ 5.6	manure and N/P/K
S-5					Bromer silt loam,
0–10	< 12	–	31	–	fallow, none
30–40	< 12	–	31	+ 5.8	

*Soil samples taken 6/06/87.

minimized ammonia volatilization and likewise the resulting positive isotopic fractionation of the residual fertilizer nitrogen. To determine if this actually happened, soil samples were taken for analysis of concentrations and isotopic compositions of leachable nitrate and ammonia (Table 2).

Soil samples S-1, S-2, and S-3 were taken from fields that had been fertilized within the previous 1–5 weeks. The low ammonia concentrations in samples S-1 and S-3 ($< 12\,\mu g\,g^{-1}$) indicate that nitrification of the fertilizer had gone to completion. The very light $\delta^{15}N$ values (+0.1 and −4‰, respectively) probably closely reflect the original isotopic compositions of the fertilizers, therefore suggesting that little ^{15}N enrichment had occurred through ammonia volatilization. The relatively large amount of ammonia remaining in sample S-2 ($109\,\mu g\,g^{-1}$) indicates that nitrification was incomplete. The $\delta^{15}N$ values of the ammonia and nitrate (+15.1 and −9.8‰, respectively) show that ^{14}N was preferentially incorporated into the first-formed nitrate, thus enriching the residual ammonia in ^{15}N. However, apparently little volatile loss of ammonia had taken place prior to sampling because the combined isotopic composition of the sampled ammonia and nitrate is approximately −0.81‰.

Soil sample S-4 was taken from a large vegetable garden that had been fertilized with manure and an N/P/K mixture. The heavier ^{15}N values (+8.8 and +5.6‰) reflect the nitrate derived from manure. Sample S-5 was taken from a fallow, but tilled field. Anhydrous ammonia had been applied to the field during the previous spring. Evidently, in the following year, the fertilizer nitrogen had been almost completely removed from the soil by plant use and/or flushing. The nitrate $\delta^{15}N$ value of +5.8‰ is within the typical range for mineralized soil organic nitrogen (i.e., +4 to +9‰; Heaton, 1986). Much of the remaining $31\,\mu g\,g^{-1}$ of nitrate may be of this origin.

Mineralized soil organic nitrogen, however, is probably not an important groundwater nitrate source throughout the mantled karst region. Macropore flow displaces little of the relatively immobile water located in small pores within the soil matrix where such nitrate is derived. If some "leaching" does occur, it happens when a small amount of nitrate diffuses from the small pores to the surfaces of the macropores (Wild, 1972; Shuford et al., 1977; Thomas and Phillips, 1979).

Most importantly, samples S-1, and S-3 show that within the tilled, 0–10 cm layer there was an abundant supply of isotopically light ($\delta^{15}N$ < +1‰) fertilizer-derived nitrate available for flushing to the aquifer. The concentrations and isotopic compositions of nitrate which could have been flushed from the fields were generally well within the range of those which would have been required to bring about the changes observed from September to May. For instance, only $62\,mg\,l^{-1}$ of nitrate with $\delta^{15}N$ = −0.54‰ would have been needed in well GW-13, assuming that a third of the water in the well was newly flushed from the surface, to have changed the nitrate concentration from 35 to $44\,mg\,l^{-1}$ and the $\delta^{15}N$ value from +8.4 to +4.2‰. The influx of fertilizer-derived nitrate resulted in no major increase of groundwater nitrate concentration in the sampled wells (Table 1), although fourteen wells did show small increases (averaging 5.6 mg l^{-1}). Three wells showed decreases, and three were unchanged. There is no correlation between the September–May changes in nitrate concentration and those in $\delta^{15}N$ value.

Like the September samples, the May samples can be categorized into either of two isotopic ranges. Eleven of the samples, with $\delta^{15}N$ values ranging from + 8.0 to + 14.7‰, were still composed of predominantly waste-derived nitrate, despite having had a larger fertilizer-derived component than when previously sampled.

The remaining nine samples probably had a fertilizer-derived component nearly equal to that of waste-derived nitrate. The $\delta^{15}N$ values ranged from + 3.6 to + 5.8‰. In samples GW-4, 11, 13, and 16 (all with $\delta^{15}N < +4.2$‰), fertilizer-derived nitrate may have been predominant.

CONCLUSIONS

Nitrate $\delta^{15}N$ values indicate that animal wastes, probably septic tank effluents, are the predominant source of groundwater nitrate in a mantled portion of a karst aquifer in southern Indiana. In the spring, when inorganic fertilizers are applied to the fields, $\delta^{15}N$ values reflect an influx of fertilizer-derived nitrate. However, this influx does not result in a large regional increase in groundwater nitrate concentration and in only a few wells does fertilizer-derived nitrate become even slightly predominant over waste-derived nitrate. Substantial influx of fertilizer-derived nitrate to the aquifer can continue only as long as an abundant supply of such nitrate exists in the tilled soil layer. As this reservoir is depleted by plant intake and loss to the aquifer, the impact of fertilizer-derived nitrate on the isotopic composition of groundwater nitrate is correspondingly diminished. The rapid response of groundwater nitrate $\delta^{15}N$ values to fertilization supports the theory that macropore flow is responsible for flushing nitrate derived from both of the above sources through the clayey mantle. Storm runoff entering macropores located beneath the tilled soil layer probably flushes nitrate from fertilized fields. Macropores draining septic tank filter fields may provide a nearly continuous flow of effluent to the aquifer.

The $\delta^{15}N$ results of this study suggest that fertilizers can be transported by macropore flow. Other agricutural chemicals (pesticides and herbicides) might be flushed to the groundwater in the same manner. This will be the subject of further research.

REFERENCES

Aley, T., 1977. A model for relating land use and groundwater quality in southern Missouri. In: R.R. Dilamarter and S.C. Csallany (Editors), Hydrologic Problems in Karst Regions. Western Kentucky Univ., Bowling Green, KY., pp. 323–332.

Bremner, J.M. and Edwards, A.P., 1965. Determination and isotope-ratio analysis of different forms of nitrogen in soils: I. Apparatus and procedure for distillation and determination of ammonium. Soil Sci. Soc. Am., Proc., 29: 504–507.

Bremner, J.M. and Keeney, D.R., 1965. Steam distillation methods for determination of ammonium, nitrate and nitrite. Geochim. Cosmochim. Acta., 32: 485–495.

Cline, J.D., 1973. Denitrification and isotopic fractionation in two contrasting marine environments: the eastern tropical North Pacific Ocean and the Cariaco Trench. Ph.D. Thesis, Univ. of California, Los Angeles, Calif., 270 pp.

Fenn, L.B. and Kissel, D.E., 1973. Ammonia volatilization from surface applications of ammonium compounds on calcareous soils: I. General theory. Soil Sci. Soc. Am., Proc., 37: 855–859.

Flipse, W.J., Jr. and Bonner, F.T., 1985. Nitrogen-isotope ratios of nitrate in ground water under fertilized fields, Long Island, New York. Ground Water, 23: 59–67.

Gormly, J.R. and Spalding, R.F., 1979. Sources and concentrations of nitrate-nitrogen in groundwater of the central Platte region, Nebraska. Ground Water, 17: 291–301.

Heaton, T.H.E., 1986. Isotopic studies of nitrogen pollution in the hydrosphere and atmosphere: a review. Chem. Geol. (Isot. Geosci. Sect.,), 59: 87–102.

Kreitler, C.W., 1975. Determining the source of nitrate in ground water by nitrogen isotope studies. Univ. of Texas, Austin, Tex., Bur. Econ. Geol., Rep. Invest., 83, 57 pp.

Kreitler, C.W., 1979. Nitrogen-isotope ratio studies of soils and groundwater nitrate from alluvial fan aquifers in Texas. J. Hydrol., 42: 147–170

Lindau, C.W. and Spalding, R.F., 1984. Major procedural discrepancies in soil extracted nitrate levels and nitrogen isotopic values. Ground Water, 22: 273–278.

Miller, J.T., Higbee, H.W., Wiancko, A.T. and Waggoner, M.E., 1939. Soil Survey of Washington County, Indiana. U.S. Dep. Agric., Bur. Chem. Soils, 65 pp.

Miyake, Y. and Wada, E., 1967. The abundance ratio of $^{15}N/^{14}N$ in marine environments. Rec. Oceanogr. Works Jpn., 9: 37–53.

Palmer, M.V. and Palmer, A.N., 1975. Landform development in the Mitchell Plain of southern Indiana: origin of a partially karsted plain. Z. Geomorphol., N.F., 19: 1–39.

Quinlan, J.F. and Aley, T., 1987. Discussion of "A new approach to the disposal of solid waste on land" by R.C. Heath and J.H. Lehr. Ground Water, 25: 615–616.

Quisenberry, V.L. and Phillips, R.E., 1976. Percolation of surface-applied water in the field. Soil Sci. Soc. Am., J., 40: 484–489.

Quisenberry, V.L. and Phillips, R.E., 1978. Displacement of soil water by simulated rainfall. Soil Sci. Soc. Am., J., 42: 675–679.

Ruhe, R.V., 1975. Geohydrology of karst terrain Lost River watershed southern Indiana. Indiana Univ. Water Resour. Res. Center, Rep. Invest., 7, 91 pp.

Shuford, J.W., Fritton, D.D. and Baker, D.E., 1977. Nitrate-nitrogen and chloride movement through undisturbed field soil. J. Environ. Qual., 6: 255–259.

Spalding, R.F., Exner, M.E., Lindau, C.W. and Eaton, D.W., 1982. Investigation of sources of groundwater nitrate contamination in the Burbank–Wallula area of Washington, U.S.A. J. Hydrol., 58: 307–324.

Sunderman, J.A., 1968. Geology and mineral resources of Washington County, Indiana. Ind. Dep. Nat. Resour., Geol. Surv. Bull., 39, 90 pp.

Thomas, G.W. and Phillips, R.E., 1979. Consequences of water movement in macropores. J. Environ. Qual., 8: 149–152.

Wells, E.R., 1988. Use of nitrogen isotope ratios to determine the source of groundwater nitrate in a mantled karst aquifer, south-central Indiana. M.S. Thesis, Indiana Univ., Bloomington, Ind. (in progress).

Wild, A., 1972. Nitrate leaching under bare fallow at a site in northern Nigeria. J. Soil Sci., 23: 315–324.

Wingard, R.C., Jr., 1984. Soil Survey of Orange County, Indiana. U.S. Dep. Agric., Soil Conserv. Ser., 124 pp.

Wolterink, T.J., Williamson, H.J., Jones, D.C., Grimshaw, T.W. and Holland, W.F., 1979. Identifying sources of subsurface nitrate pollution with stable nitrogen isotopes. U.S. Environ. Prot. Agency, EPA-600/4-79-050, 151 pp.